Physics 710 February 19, 2010

Problem Set 12

Problem 11.2.2: Since $I = T^\dagger(\epsilon)T(\epsilon) = [I+(i\epsilon/\hbar)G^\dagger+\mathcal{O}(\epsilon^2)]\cdot[I-(i\epsilon/\hbar)G+\mathcal{O}(\epsilon^2)] = I + (i\epsilon/\hbar)(G^\dagger - G) + \mathcal{O}(\epsilon^2)$, therefore

$$0 = \frac{i\epsilon}{\hbar}(G^\dagger - G) + \mathcal{O}(\epsilon^2).$$

Since this is true for all ϵ, it implies that $G^\dagger - G = 0$.

Problem 11.4.1: If $[\Pi, H] = 0$, and $\Pi|\psi(0)\rangle = \pm|\psi(0)\rangle$, then $|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$, and

$$\begin{aligned}
\Pi\,|\psi(t)\rangle &= \Pi\, e^{-iHt/\hbar}|\psi(0)\rangle \;=\; e^{-iHt/\hbar}\,\Pi\,|\psi(0)\rangle \qquad \text{since } [\Pi, H] = 0 \\
&= e^{-iHt/\hbar}(\pm)|\psi(0)\rangle \;=\; \pm e^{-iHt/\hbar}|\psi(0)\rangle \;=\; \pm|\psi(t)\rangle.
\end{aligned}$$

Problem 11.4.3: Parity takes $\Pi : \vec{x} \to -\vec{x}$ and $\Pi : \vec{p} \to -\vec{p}$. So for angular momentum, $\vec{L} = \vec{x}\times\vec{p}$, $\Pi\,\vec{L} = (-\vec{x})\times(-\vec{p}) = \vec{x}\times\vec{p} = +\vec{L}$. So if in the reaction $\vec{L}$ is parallel to $\vec{p}$, then in the parity-reversed state $\vec{L} \to \vec{L}$ and $\vec{p} \to -\vec{p}$, so $\vec{L}$ is *anti-parallel* to $\vec{p}$. (Note: spin is just a kind of angular momentum.) Therefore the reaction is *not* parity-preserving since it takes an arbitrary parity initial state, but always gives a final state of a definite parity.